Recap-1
The Compilers' Front End

Regular Expression (Regex)

- Regex describes a language
- Primitive regex
  - $\emptyset, \epsilon, a$
- Given two regex: r_1, r_2, the following are regex
  - $r_1|r_2$
  - $r_1 r_2$
  - r_1^*
  - (r_1)
- **Example:** $(a|b)^*c$

Regex to NFA (DFA)

- Primitive regex
  - $L(\emptyset) = \emptyset;\ \ L(\epsilon) = \{\epsilon\};\ \ L(a) = \{a\}$
- Given two regex: r_1, r_2, the following are regex
  - $L(r_1|r_2) = L(r_1) \cup L(r_2)$
  - $L(r_1 r_2) = L(r_1)L(r_2)$
  - $L(r_1^*) = (L(r_1))^*$
  - $L((r_1)) = L(r_1)$

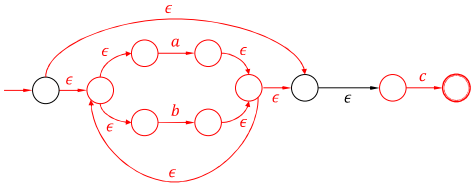
PART I: Regex → NFA → (Min) DFA

Language by Regex

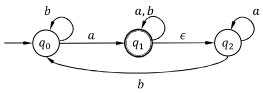
- The language represented by regex are defined below
- Primitive regex
  - $L(\emptyset) = \emptyset;\ \ L(\epsilon) = \{\epsilon\}\ ; L(a) = \{a\}$
- Given two regex: r_1, r_2, the following are regex
  - $L(r_1|r_2) = L(r_1) \cup L(r_2)$
  - $L(r_1 r_2) = L(r_1)L(r_2)$
  - $L(r_1^*) = (L(r_1))^*$
  - $L((r_1)) = L(r_1)$

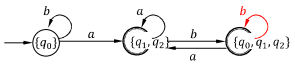
Example

- Build the NFA for the regex:$(a|b)^*c$
- $L\big((a|b)^*c\big) = \big(L(a|b)\big)^* L(c) = \big(L(a) \cup L(b)\big)^* L(c)$



From NFA to DFA





- **Subset Construction**
- A subset of NFA states is a DFA state

PART II: CFG and Parsing

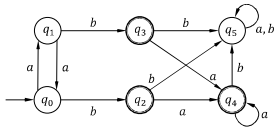
Exercises

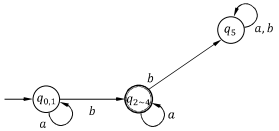
$S \to aSb \mid \epsilon$ is the grammar for $a^n b^n$

- Write a context-free grammar for the following languages
  - 0^*1^*
  - 0^n1^{2n}
  - L = { w | w is a string of balanced parentheses }
  - $a^i b^j c^k$ where $i = j$ or $j = k$
  - $a^i b^j c^k$ where $i \neq j$ or $j \neq k$

DFA Minimization

- Minimizing DFA can improve the efficiency of computation





Step 1	$\{q_0, q_1, q_5\}, \{q_2, q_3, q_4\}$	Distinguish final and non-final states
Step 2	$\{q_0, q_1\}, \{q_5\}, \{q_2, q_3, q_4\}$	$\delta(q_0, a/b)$ and $\delta(q_1, a/b)$ are in the same set, but $\delta(q_5, a/b)$ not
Step 3	$\{q_0, q_1\}, \{q_5\}, \{q_2, q_3, q_4\}$	The result does not change and the algorithm completes

Context Free Grammar (CFG)

- A context-free grammar is a tuple $G = (N, T, S, P)$
  - N: a finite set of non-terminals
  - T: a finite set of terminals, such that $N \cap T = \emptyset$
  - $S \in N$: start non-terminals
  - P: production rules in the form of $A \to a$, where $A \in N$ and $a \in (N \cup T)^*$

assignment → identifier = expression
expression → term + term
term → identifier
term → identifier * number

assignment → identifier = expression
expression → term + term
term → identifier
| identifier * number

Building a Parser in Practice

- Each non-terminal has a procedure or function for parsing
- => Recursive-Descent Parser

- **Problem 1**: A **left-recursive** grammar can cause **infinite loops**
  - when expanding a non-terminal, we may find itself and expand it again
- **Problem 2**: Backtracking may be necessary
  - when one derivation does not work, we may try others

Eliminating Left-Recursion

- A grammar is left-recursive if it has a non-terminal A such that there is a derivation $A \Rightarrow^{+} A\alpha$
- **Example:** $A \to A\alpha \mid \beta$ is left-recursive, can be transformed into

$$A \to \beta A'$$
$$A' \to \alpha A' \mid \epsilon$$

$$A \to A\alpha_1 \mid A\alpha_2 \mid \cdots \mid A\alpha_m \mid \beta_1 \mid \beta_2 \mid \cdots \mid \beta_n$$

$$A \to \beta_1 A' \mid \beta_2 A' \mid \cdots \mid \beta_n A'$$
$$A' \to \alpha_1 A' \mid \alpha_2 A' \mid \cdots \mid \alpha_m A' \mid \epsilon$$

Building a Parser in Practice

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Predictive Parsing

- Predictive parsers are recursive descent parser w/o backtracking
- LL(1)
  - L: scanning input from left to right
  - L: leftmost derivation
  - 1: Using one input symbol of lookahead at each step

Predictive Parsing

- Predi backtracking
- LL(1)
  - L:
  - L:
  - 1:

NON-TERMINAL	INPUT SYMBOL					
	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'		$T' \to \epsilon$	$T' \to *FT'$		$T' \to \epsilon$	$T' \to \epsilon$
F	$F \to$ id			$F \to (E)$		

- LL(1) grammar (Not ambiguous! Not left-recursive!)
  - Rich enough to cover most programming constructs
  - To build the predictive table, let's define $\text{FIRST}(\alpha)$; $\text{FOLLOW}(\alpha)$

First() and Follow()

- $\text{FIRST}(\alpha)$:
  - A set of terminals that α may start with
- $\text{FOLLOW}(\alpha)$:
  - A set of terminals that can appear immediately to the right of α

First()

- **Example**: $S \to c\ A\ b;\ \ A \to a\ b \mid a$
- $\text{FIRST}(a) = \{a\}$
- $\text{FIRST}(b) = \{b\}$
- $\text{FIRST}(c) = \{c\}$
- $\text{FIRST}(S) = \{c\}$
- $\text{FIRST}(A) = \{a\}$

Follow()

- FOLLOW(α):
  - A set of terminals that can appear immediately to the right of α
  - $\$ \in \text{FOLLOW}(S)$, where $ is string's end marker, S the start non-terminal

Follow()

- **Example**: $S \rightarrow c\ A\ b;\ A \rightarrow a\ b \mid a$
- $\text{FOLLOW}(S) = \{\$\}$
- $\text{FOLLOW}(A) = \{b\}$

- **Example**: $S \rightarrow c\ A\ A;\ A \rightarrow a\ b \mid a$
- $\text{FOLLOW}(S) = \{\$\}$
- $\text{FOLLOW}(A) \supseteq \text{FIRST}(A) \backslash \{\epsilon\} = \{a\}$
- $\text{FOLLOW}(A) \supseteq \text{FOLLOW}(S) = \{\$\}$

Predictive Parsing Table

- To build a parsing table $M[A, a]$, for each $A \rightarrow \alpha$
  - $\forall a \in \text{FIRST}(\alpha): M[A, a] = A \rightarrow \alpha$
  - $\epsilon \in \text{FIRST}(\alpha) \Rightarrow \ \forall b \in \text{FOLLOW}(A): M[A, b] = A \rightarrow \alpha$

E	$\rightarrow$	$T\ E'$
E'	$\rightarrow$	$+\ T\ E' \mid \epsilon$
T	$\rightarrow$	$F\ T'$
T'	$\rightarrow$	$*\ F\ T' \mid \epsilon$
F	$\rightarrow$	$(\ E\ ) \mid \textbf{id}$

Non-terminal	Input Symbol					
	id	+	*	(	)	$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow +TE'$			$E' \rightarrow \epsilon$	$E' \rightarrow \epsilon$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow \epsilon$	$T' \rightarrow *FT'$		$T' \rightarrow \epsilon$	$T' \rightarrow \epsilon$
F	$F \rightarrow \textbf{id}$			$F \rightarrow (E)$		

- **Choose the production rule as per the table; empty means error**

Building a Parser in Practice

- Each non-terminal has a procedure or function for parsing
- => Recursive-Descent Parser

- **Problem 1**: A **left-recursive** grammar can cause **infinite loops**
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PART III: IR Generation

Three-Address Code

- do i = i + 1; while (a[i + 2] < v);

```
L:   t1 = i + 1
     i = t1
     t2 = i + 2
     t3 = a [ t2 ]
     if t3 < v goto L
```

Symbolic Labels

```
100:  t1 = i + 1
101:  i = t1
102:  t2 = i + 2
103:  t3 = a [ t2 ]
104:  if t3 < v goto 100
```

Numeric Labels

- Implementation methods: *quadruples*, *triples*, etc.

Static Single-Assignment

- **Feature 1**: Every variable has only one definition
- **Feature 2**: Using φ to merge definitions from multi paths
- => Direct def-use chains

```
if ( flag ) x = -1; else x = 1;
y = x * a;
```

```
if ( flag ) x1 = -1; else x2 = 1;
x3 = φ (x1, x2);
y = x3 * a
```


Dominance Relations

- A **dom** B
  - if all paths from Entry to B goes through A
- A **post-dom** B
  - if all paths from B to Exit goes through A

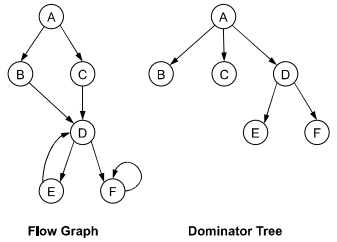
- Strict (post-)dominance – A (post-)**dom** B but A ≠ B

- Immediate dominance – A strict-dom B, but there's no C, such that A strict-dom C, C strict-dom B

Dominator Tree

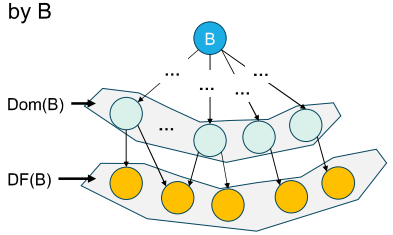
- Almost linear time to build a dominator tree.
  - Node: Block
  - Edge: Immediate dom relation

- Why is it a tree?



Dominance Frontier

- DF(B) = { … } for the block B
  - The immediate successors of the blocks dominated by B
  - Not strictly dominated by B



Dominance Frontier

- DF(B) = { … } for the block B
  - The immediate successors of the blocks dominated by B
  - Not strictly dominated by B

- DF(𝔹) = { … } for a set of blocks 𝔹
  - $DF(\mathbb{B}) = \cup_{B \in \mathbb{B}} DF(B)$

Iterated Dominance Frontier

- Iterated DF of a block set 𝔹
  - $DF_1 = DF(\mathbb{B})$; $\mathbb{B} = \mathbb{B} \cup DF_1$
  - $DF_2 = DF(\mathbb{B})$; $\mathbb{B} = \mathbb{B} \cup DF_2$
  - ……
  - until fixed point! (i.e., $DF_n = DF_{n-1}$)